

Slipstream

Instant game launch for certified casino bundles

FEG Hackathon 2026 · Challenge 03: Game Load Time

Team Equinox

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The problem

- A game is a 56 MB certified bundle. On a normal mobile connection it takes 5.4 seconds before the player can press Play, and another 45 MB arrives while they play.
- Because every tap costs six to eight seconds, players stop trying new games. They stay with a few they know, and the rest of the catalogue goes unused.
- We cannot make the bundle smaller. It is certified, so it cannot be changed. It also cannot move to a separate CDN domain, because a service worker cannot see across domains.
- And we cannot simply download it for everyone: about 95% of people who open the lobby never open a game.

So the only thing left to change is when the bytes are downloaded, and for whom.

What we built

A service worker downloads the game in the background while the player is still looking at the lobby, in three steps:

1. Play screen	6.8 MB	the 27 files needed before Play can be pressed	everyone
2. Sound	10.7 MB	the audio the game loads at start-up	everyone, after step 1
3. Everything else	38.8 MB	animations, pay table, bonus round music	players who have played before

- All 30 games in the lobby use the same bundle, so one download makes every game instant. There is no guessing which game to prepare for, and no wasted data.
- The bundle itself is never touched. We only store and serve the same files, so the game stays exactly as certified and nothing needs approving again.

Results

Time for a game to finish loading completely, with nothing left to download.

Connection	Speed	Before	After	Faster
Slow 3G	2 Mbps	3 min 23 s	0.8 s	254x
3G	5 Mbps	1 min 21 s	0.8 s	102x
Throttled test (measured)	8 Mbps	51 s	0.8 s	64x
4G average	20 Mbps	20 s	0.8 s	25x
Good 4G	50 Mbps	8 s	0.8 s	10x
Wi-Fi / fibre	100 Mbps	4 s	0.8 s	5x

The "After" column is the same on every connection because nothing is downloaded — the game is read from disk. On fibre this saves 3 seconds; on a slow mobile connection it saves 3 minutes.

Time to the Play screen: 5.43 s before, 0.24 s after. The challenge asked for 0.5 s.

And if the player opens a game while the download is still running, it takes 2.77 s — still half the original wait. The download is already fetching the files that game needs, so it helps rather than competes with it.

Cost and impact

- Preparing the game costs about 17.5 MB per lobby visit. At €0.02 per GB that is €0.35 per 1,000 visits, and most of it is not extra: anyone who opens a game would have downloaded those bytes anyway. The genuinely extra cost is about €0.24 per 1,000 visits.
- In return, each playing session saves about 27 seconds of waiting — roughly 2.3 hours per 1,000 lobby visits.
- If one playing session is worth €0.50, this pays for itself with a 0.16% increase in sessions. Even at five times the bandwidth price it stays below 1%.
- The largest download only goes to players who are likely to use it. In the sample event data, someone with no history opens a game 4.5% of the time; someone with 10 or more past sessions, 46.7%.

Main assumptions

- Games keep sharing one bundle. If each game had its own, this would not work as well.
- Players come back. The largest download is only worth it for returning players.
- €0.02 per GB bandwidth — to be replaced with the operator's real rate.
- The behaviour data is 89 players over one month, so we would repeat it on full traffic.